

Sea Level is assumed to be Y = 63 in numerous parts of Minecraft

MC-226687 Custom Worlds

Status	Resolved / Fixed
Affected	1.17 Pre-release 1, 1.17, 1.18 Pre-release 3, 1.18 Pre-release 4, 1.18.1, 22w12a (+5 more)
Fixed in	24w33a
Reported	2021-05-28

STEPS TO REPRODUCE

1. Launch Minecraft and create a new Singleplayer Creative world with default settings.
2. Enter the world, then use the in-game "Esc menu -> Save and Quit to Title".
3. Select the world and open "Singleplayer -> [world] -> Edit -> Open World Folder".
4. Copy the attached `TallOceansTest` datapack into the world's `datapacks` folder.
5. Reopen the world and travel to an ocean generated by the datapack.
6. Wait at least a few minutes for natural mob spawning to occur.
7. Observe that the ocean's water level is above Y 63, while the affected aquatic mobs do not spawn there as expected; other sea-level-dependent behavior, such as blue ice generation or ocean monument water placement, may also be inconsistent with the configured sea level.
8. Repeat steps 1-5 using the attached `SnowTestUnder63` datapack instead.
9. Observe that terrain and temperature-related behavior, including snow placement, still uses the vanilla Y 63/Y 64 assumptions even though the datapack configures a lower sea level.

OBSERVED BEHAVIOUR

When the chunk generator is configured with a sea level other than Y 63, many systems continue using the value returned by `Level.getSeaLevel` as though it were the generator's actual sea level. Because this value is not synchronized with the chunk generator, aquatic mob spawning becomes incorrect. Dolphins, drowned, turtles, and squid may fail to spawn in ocean water when the configured sea level is above Y 63, while axolotls and glow squids may only spawn when the ocean floor is below Y 63. Ocelot spawn obstruction checks also use the incorrect level, causing spawning to be incorrectly allowed below an elevated sea level and incorrectly obstructed when the configured sea level is below Y 63. Phantom anchor points are clamped to Y 64 instead of to one block above the configured sea level, and phantom spawning rates or locations change incorrectly when the sea level differs from Y 63. Bats fail to spawn in caves below an elevated sea level and can spawn in areas where they should not when the configured sea level is lower. Drowned spawning uses a fixed effective threshold of Y 58 rather than five blocks below the configured sea level. The blue ice feature fails to generate in water bodies above Y 62, regardless of the configured sea level. Ocean monuments use hardcoded Y 64 behavior in `OceanMonumentPieces.postProcess` and related water-box generation, causing their placement and filling to disregard the chunk generator's sea level and ocean-floor height. Ocean ruins similarly place air or water incorrectly around drowned data markers because their block replacement logic uses the unsynchronized sea-level value. Amphibious pathfinding for axolotls and turtles applies the shallow-water cost adjustment relative to the incorrect sea level, which can produce inaccurate paths when water is generated at different heights. Biome temperature adjustment also remains based on Y 64, causing temperature and snow behavior to be incorrect when the configured sea level is changed, particularly in the `SnowTestUnder63` datapack. These problems can be reproduced with the attached `TallOceansTest` and `SnowTestUnder63` datapacks and indicate that code relying on `Level.getSeaLevel` is not receiving the chunk generator's configured sea level. The server-side level should obtain this value from the active chunk generator, and the corresponding sea-level information should also be made available to clients where client-side calculations require it.

EXPECTED BEHAVIOUR

All sea-level-dependent systems should use the chunk generator's configured sea level rather than assuming Y=63. Mob spawning, obstruction checks, phantom behavior, pathfinding, blue ice and ocean feature generation, monument and ruin water placement, and biome temperature adjustments should remain correct when sea level is raised or lowered, with water, air, height thresholds, and environmental behavior consistently aligned to the

configured value.

ENVIRONMENT

Minecraft Java Edition Versions: 1.17 Pre-release 1, 1.17, 1.18 Pre-release 3, 1.18 Pre-release 4, 1.18.1, 22w12a, 1.19.2, 1.19.3, 23w07a, 1.20.2, 23w45a (includes snapshots)

ORIGINAL DESCRIPTION

The bugs listed here all stem from the bugged method in the screenshot(`Level.getSeaLevel`). Several bug reports were NOT made because fixing this singular bugged method would fix all the bugs listed.

Since 1.16's datapack system has been introduced, sea level has become a configurable argument in the chunk generator.

But there's one issue, a lot of places still assume that sea level is 63 when this is no longer the case when changing the sea level in the chunk generator.

I've attached the `TallOceansTest & SnowTestUnder63` Datapack to recreate these issues listed.

This source of the issue can be narrowed down to `Level.getSeaLevel` method and usages of it as shown here:

This means all usages of this method, are possibly "broken" when this value is not changed together with Chunk Generator's sea level.

My solution:

In `ServerLevel`, override this method and call: `this.chunkSource.getGenerator.getSeaLevel` and on the client you'd most likely need some sort of packet containing the sea level of the current chunk generator/`ServerLevel`.

Here are a list of issues that may occur when changing sea level:

1. Dolphins, Drowned, Turtles, & Squid spawning does not occur in oceans in water when sea level is above Y 63.

No real good way to visualize this. But I've attached a datapack which increases both ocean depth & and the chunk generator's sea level to show this issue. All of the code for these can be found in the respective Mob's spawn rules.

2. Axolotls & Glowsquid spawning only occurs when the Ocean Floor Heightmap is under Y 63 as seen here in the `WaterAnimal` class:

```
public static boolean checkUndergroundWaterCreatureSpawnRules(EntityType<? extends LivingEntity> entityType,
LevelAccessor world, MobSpawnType spawnReason, BlockPos pos, Random random) {
return pos.getY() < world.getSeaLevel() && pos.getY() < world.getHeight(Heightmap.Types.OCEAN_FLOOR,
pos.getX(), pos.getZ());
}
```

3. Ocelot spawning is not obstructed under the chunk generator sea level(when above 63) and ALWAYS obstructed when Chunk Generator sea level/terrain is under 63.

As seen here in the `Ocelot` class.

```
public boolean checkSpawnObstruction(LevelReader world) {
if (world.isUnobst
```

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-226687>